Exercise 5 NULL Functions

Q1) Show all employees with their salary. If salary is NULL, display 0.

1) Table 1: Employees

employee_id	name	department	Salary
1	Alice	HR	5000
2	Bob	IT	NULL
3	Charlie	NULL	7000
4	Dana	Finance	NULL

```
SELECT employee_id AS employee_id
       name AS name
       COALESCE(TRY_CAST(Salary AS DOUBLE),0) AS
                salary_with_default
FROM Employees;
```

--employee_id	name	salary_with_default
1	Alice	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Question 2

Show employee name with their department. If department is NULL, show "Not Assigned".

- Expected columns: employee_id, name, department_name

```
SELECT employee_id,
       name,
       IFNULL( department, 'Not Assigned') AS department_name
FROM Employees;
```

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance

Table 2: Orders

order_id	customer_id	delivery_date
101	201	2024-12-01
102	202	NULL
103	NULL	2024-12-03

Q3: Find orders with Null customer_id using ISNULL()
Expected Columns: order_id, customer_id

```
SELECT order_id, customer_id
FROM Orders
WHERE customer_id IS NULL;
```

order_id	customer_id
103	null

Question 4

Show all orders. If delivery_date is NULL, show 'Pending'.
Expected columns: order_id, customer_id, delivery_status

```
SELECT order_id, customer_id,
        IFNULL(delivery_date, 'Pending') AS delivery_status
FROM Orders;
```

order_id	customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	null	2024-12-03

Table 3: Students

student_id	name	grade
1	Ethan	85
2	Maya	NULL
3	Olivia	90

Question 5

Show all students and their grades. Replace NULL with 0.
Expected columns: student_id, name, final_grade

```
SELECT student_id, name,
        IFNULL(grade, 0) AS final_grade
FROM Students;
```

student_id	name	final_grade
1	Ethan	85
2	Maya	0
3	Olivia	90

Question 6

Count students who havent been graded

Expecte columns: not_grade_count

```
SELECT COUNT(*) AS not_graded_count
FROM students
WHERE grade IS NULL;
```

not_graded_count
1

Table 4: Products

Product_id	name	Price	discount
501	Keyboard	25	NULL
502	Mouse	15	5
503	Monitor	100	NULL

Question 7

Show product name, price, and final price after discount (assume 0 if discount is NULL).

Expected columns: product_id, name, final_price

```
SELECT product_id, name
        price - IFNULL(discount, 0) AS final_price
FROM Products;
```

product_id	name	final_price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Table 5 : Customers

customer_id	name	email
1	Linda	NULL
2	Joseph	joseph@mail.com
3	Nia	Null

Question 8

Count how many customers have no email.

Expected columns : missing_email_count

```
SELECT COUNT(*) AS missing_email_count
FROM Customers
WHERE email IS NULL;
```

missing_email_count
2

Question 9

Show all customer with email. If NULL, display "No Email".

Expected columns : count_id, name, email_display

```
SELECT customer_id, name,
        IFNULL(email, 'No Email') AS email_display
FROM Customers;
```

customer_id	name	email_display
1	Linda	No Email
2	Joseph	joseph@mail.com
3	Nia	No Email

Table 6: Payments

payment_id	method	status
301	Credit	NULL
302	Paypal	Success
303	NULL	Failed

Question 10

Show payment details with method replaced by "Unknown" if NULL.

Expect columns: payment_id, method_display, status

```
SELECT payment_id,
       IFNULL(method, 'Unknown') AS method_display,
       status
FROM payments;
```

payment_id	method_display	status
301	Credit	null
302	Paypal	Success
303	Unknown	Failed

Question 11

Table 7: Inventory

item_id	item_name	quantity
1	Pen	NULL
2	Notebook	150
3	Eraser	NULL

Show items and their quantity (0 if NULL).
Expected Column: item_id, item_name, quantity_checked

```
SELECT item_id, item_name,
        IFNULL(quantity, 0) AS quantity_checked
FROM Inventory;
```

item_id	item_name	quantity_checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Table 8: Employees_Extra

emp_id	bonus	COMMISSION
1	NULL	300
2	100	NULL
3	NULL	NULL

Question 12

Show employee ID and the first available value among bonus or commission.
Expected Column: emp_id, first_available_reward

```
SELECT emp_id,
        COALESCE(bonus, commission) AS first_available_reward
FROM Employees_Extra;
```

emp_id	first_available_reward
1	300
2	100
3	null

Table 9: Classes

class_id	subject	room
11	Math	NULL
12	Science	Lab A
13	English	NULL

Question 13

Count classes that don't have a room assigned.

Epected columns: no_room_count

```
SELECT COUNT(*) AS no_room_count
FROM Classes
WHERE room IS NULL;
```

no_room_count
2

Table 10: Attendance

student_id	data	status
1	2025-04-01	NULL
2	2025-04-01	Present
3	2025-04-01	Absent

Question 14

Show attendance records with status. Replace NULL with "Not Marked".

Expected columns: student_id, attendance_status, date

```
SELECT student_id, date,
        IFNULL(Status, 'Not Marked') AS attendance_status
FROM Attendance;
```

student_id	date	attendance_status
1	2025-04-01	Not Marked
2	2025-04-01	Present
3	2025-04-01	Absent

Table 11: Bank_Accounts

account_id	account_type	balance
A1	Savings	NULL
A2	Current	5000
A3	NULL	2000

Question 15

Show account ID, account_type (or 'Unknown'), and balance (or 0).

Expected columns: account_id, type_display, balance_checked.

```
SELECT account_id,
        IFNULL(account_type, 'Unknown') AS type_display,
        IFNULL(balance, 0) AS balance_checked
FROM Bank_Accounts;
```

account_id	type_display	balance_checked
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Table 12 : Projects

project_id	title	start_date	end_date
1	Website Revamp	2025-01-10	NULL
2	Mobile App	NULL	2025-06-01
3	Data Migration	NULL	NULL

Question 16

Show all projects with a start date. If start_date is NULL, display 'TBD'.
Expected columns : project_id, title, start_display

```
SELECT project_id, title,
        IFNULL(start_date, 'TBD') AS start_display
FROM Projects;
```

Project_id	title	start_display
1	Website Revamp	2025-01-10
2	Mobile App	TBD
3	Data Migration	TBD

Table 13: Reviews

review_id	product_id	comment	rating
1	501	Great product	4
2	502	NULL	NULL
3	503	Works fine	3

Question 17

Display reviews showing comment (or 'No Comment') and rating (or 0).

Expected columns: review_id, product_id, comment_display, rating_display

```
SELECT review_id, product_id,
        IFNULL(comment, 'No Comment') AS comment_display,
        IFNULL(rating, 0) AS rating_display
FROM Reviews;
```

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No Comment	0
3	503	Works fine	3

Table 14: Suppliers

supplier_id	name	phone	alt_phone
1	Global Goods	NULL	123456789
2	Best Suppliers	987654321	NULL
3	ValueSource	NULL	NULL

Question 18

Show the supplier contact number. Use COALESCE(phone, alt_phone, 'No Contact').

Expected columns: supplier_id, name, contact_number

```
SELECT supplier_id, name,
        COALESCE(phone, alt_phone, 'No Contact') AS
        contact_number
FROM Suppliers;
```

supplier_id	name	contact_number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	No Contact

Table 15: User_Settings

user_id	theme	language	timezone
1	NULL	English	NULL
2	Dark	NULL	UTC+1
3	NULL	NULL	NULL

Question 19

Show all users and their preferences. Replace all NULLs with defaults:

* Theme -> "Light"
* Language -> "English"
* Timezone -> "UTC"

Expected Columns: user_id, theme_set, language_set, timezone_set

```
SELECT user_id,
        IFNULL(theme, 'Light') AS theme_set,
        IFNULL(language, 'English') AS language_set,
        IFNULL(timezone, 'UTC') AS timezone_set
FROM User_Settings;
```

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Table 16: Maintenance

record_id	machine_id	issue	technician
1	M101	Overheating	NULL
2	M102	NULL	NULL
3	M103	Jammed	Alex

Question 20

Show maintenance log with:

- issue -> default to "Unknown Issue"
- technician -> default to "Not Assigned"

Expected columns: record_id, machine_id, issue_log, technician_name

```
SELECT record_id, machine_id,
        IFNULL(issue, 'Unknown Issue') AS issue_log,
        IFNULL(technician, 'Not Assigned') AS technician_name
FROM Maintenance;
```

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown Issue	Not Assigned
3	M103	Jammed	Alex